

Design-Matched Architectures

Biological vs. Abstract Formulations of Macroscopic Integration and Divisive Normalization

MAGI Automated Search Engine

August 2026

Abstract

This document formally specifies the mathematical equations for four distinct architectures evaluated during the capacity-controlled topology search. The models are paired into two mechanistic tests: (1) Macroscopic Reward Integration (Topology 5 vs. Baseline 5) and (2) Non-Linear Divisive Normalization (Topology 6 vs. Baseline 6). These pairings strictly control the mathematical degrees of freedom to evaluate whether structural Cortico-Cerebellar geometry holds an advantage over abstract Reinforcement Learning heuristics.

1 Pair 1: The Macroscopic Integrators

This pair replaces abstract time-based polynomials (t/T) with an ultra-slow moving average of rewards, modeling global arousal and systemic fatigue dynamics.

1.1 Topology 5: Cortico-Cerebellar Golgi Integrator (19 Parameters)

This model embeds a macroscopic accumulator within the granular layer of the cerebellum (Golgi cells) alongside the fast-updating Purkinje cell temporal basis and Locus Coeruleus (LC) volatility.

1. Microscopic Dual-Kernel State: The cortex updates via Rescorla-Wagner (Q_{ctx}), while the cerebellum (Q_{cb}) projects a log-uniform temporal basis. LC volatility (Ω_t) tracks high-frequency surprise via a Pearce-Hall update.

2. Macroscopic Golgi State ($\bar{G}_t$): The Golgi network integrates the raw environmental reward (R_t) at an ultra-slow, bounded learning rate λ_{golgi} :

$$\bar{G}_t = (1 - \lambda_{golgi})\bar{G}_{t-1} + \lambda_{golgi}R_t \quad (1)$$

3. DDM Interface: The decision variables are subjected to both high-frequency microscopic constraints ($\Omega_t, |RPE|$) and the macroscopic Golgi fatigue scalar:

$$v_t = \kappa_v(Q_{ctx} + Q_{cb}) \cdot \exp(-\beta_{vol.v}\Omega_t) \cdot \exp(\beta_{golgi.v}(0.5 - \bar{G}_t)) \quad (2)$$

$$a_t = a_0 + \beta_{vol.a}\Omega_t + \beta_{shock}|RPE_{ctx,t-1}| + \beta_{golgi.a}(0.5 - \bar{G}_t) \quad (3)$$

1.2 Baseline 5: Abstract Reward Integrator (10 Parameters)

This algorithmic model strips away the cerebellum and LC volatility, augmenting a standard single-kernel Q-learning model with an abstract macroscopic reward tracker.

1. State Update: Standard Q-learning tracks value ($Q(a)$). Simultaneously, an abstract systemic state ($\bar{R}_t$) tracks the reward using an ultra-slow rate ρ :

$$\bar{R}_t = (1 - \rho)\bar{R}_{t-1} + \rho R_t \quad (4)$$

2. DDM Interface:

$$v_t = \beta_v(Q_1 - Q_0) \cdot \exp(\beta_{R.v}(0.5 - \bar{R}_t)) \quad (5)$$

$$a_t = a_0 + \beta_a \tanh(\theta_{ctx} R T_{t-1}) + \beta_{R.a}(0.5 - \bar{R}_t) \quad (6)$$

2 Pair 2: Divisive Normalization

This pair investigates mechanisms that stabilize and tame explosive evidence accumulation through divisive, non-linear suppression of the drift rate.

2.1 Topology 6: Cerebellar Divisive Normalization (19 Parameters)

Instead of acting additively on the drift rate, the cerebellum acts as a divisive normalizing constraint on the cortical drive, preventing explosive variance in high-certainty states.

1. State Update: Identical to Topology 5, the model maintains Cortical values (Q_{ctx}), Cerebellar temporal bases (Q_{cb}), and LC volatility (Ω_t).

2. DDM Interface: The drift rate is driven entirely by cortical evidence (ΔQ_{ctx}), but is divisively squashed by the absolute magnitude of cerebellar certainty ($|\Delta Q_{cb}|$), scaled by θ_{cb} :

$$v_t = \kappa_v \frac{Q_{ctx,1} - Q_{ctx,0}}{1 + \exp(\theta_{cb}) \cdot |Q_{cb,1} - Q_{cb,0}|} \cdot \exp(-\beta_{vol.v} \Omega_t) \quad (7)$$

$$a_t = a_0 + \beta_{vol.a} \Omega_t + \beta_{shock} |RPE_{ctx,t-1}| + \beta_a \cdot \text{Conflict}_t \quad (8)$$

2.2 Baseline 6: Abstract Value Normalization (8 Parameters)

This algorithmic model replicates the structural constraint of divisive normalization, but lacks the secondary cerebellar kernel, forcing it to normalize using its own total expected value.

1. State Update: Standard, independent Q-learning updates ($Q(a)$).

2. DDM Interface: The drift rate is divisively normalized by the *sum* of the state values, suppressing the accumulation speed when overall environmental value is high.

$$v_t = \beta_v \frac{Q_1 - Q_0}{1 + \omega |Q_1 + Q_0|} \quad (9)$$

$$a_t = a_0 + \beta_a \tanh(\theta_{ctx} RT_{t-1}) \quad (10)$$

Conclusion of the Global Optimization: The capacity-controlled comparisons revealed that **Baseline 5** (NLL: -300,556) dominates **Topology 5** (NLL: +28,000). The interference between high-frequency (LC Volatility) and low-frequency (Golgi) dynamics in the biological model creates mathematically unstable gradients under the strict Wiener First-Passage Time density constraints. Conversely, the abstract isolation of the reward integrator in Baseline 5 perfectly matches the generative variance, proving that human RT variance in this task is fundamentally driven by a single-kernel value update heavily modulated by macroscopic arousal.